

A. Data Analysis and Insight

1. Data Overview

The dataset contains information about students' demographic background, study habits, academic behaviour, subject scores, and final exam results.

The dataset includes the following key variables:

- student_id: Unique identifier for each student
- gender: Student gender
- age: Age of the student
- parental_education: Highest education level of parents
- family_income: Household income category
- internet_access: Whether the student has internet access
- study_environment: Quality of study environment
- study_hours_per_day: Average daily study time
- attendance_rate: Percentage of class attendance
- sleep_hours: Average hours of sleep per night
- social_media_hours: Average daily time spent on social media
- assignment_completion_rate: Percentage of assignments completed
- participation_score: Level of student participation in class
- online_courses_completed: Number of online courses completed
- tutoring: Whether the student receives tutoring support
- math_score: Score achieved in mathematics
- reading_score: Score achieved in reading
- writing_score: Score achieved in writing
- science_score: Score achieved in science
- final_exam_score: Score achieved in the final exam

- previous_gpa: Student's GPA from previous academic periods
- pass_fail: Indicates whether the student passed or failed
- grade_category: Classification of overall performance

Dataset size:

- Number of rows: 10,000
- Number of columns: 23

	student_id	gender	age	parental_education	family_income	internet_access	study_environment	study_hours_per_day	attendance_rate	sleep_hours	social_media_hours	assignment_completion_rate	participati
2	S00001	Male	17	High School	Medium	Yes	Quiet	2.98	96.5	6.05	0.1		80.5
3	S00002	Female	18	High School	Low	Yes	Quiet	4.45	95.7	6.96	2.9		70.9
4	S00003	Male	17	High School	Medium	No	Quiet	3.75	76	7.02	2.4		77.6
5	S00004	Male	18	Bachelor	Medium	Yes	Quiet	2.03	72.6	6.23	3.5		63.5
6	S00005	Male	18	Bachelor	Medium	Yes	Quiet	5.14	87.3	8.54	2.1		71.8
7	S00006	Female	17	Master	Medium	Yes	Quiet	1.16	92	8.16	2.6		100
8	S00007	Male	16	High School	High	Yes	Quiet	3.33	74.2	7.34	3.9		81.3
9	S00008	Male	18	Bachelor	Low	Yes	Moderate	2.01	88	6.58	0		87.3
10	S00009	Male	17	Master	Medium	Yes	Moderate	1.22	70.4	7.96	1.1		79.6
11	S00010	Female	15	Master	Low	Yes	Quiet	1.64	95.6	7.51	1.7		81
12	S00011	Male	17	Master	Medium	Yes	Noisy	2.94	79.3	6.56	2.3		82
13	S00012	Male	15	Bachelor	Low	Yes	Moderate	3.42	83.4	7.97	0.4		76.4
14	S00013	Male	18	Master	Low	No	Moderate	4.27	93.6	7.37	1.1		88.9
15	S00014	Male	17	High School	Medium	Yes	Noisy	4.77	92.7	5.34	3.5		69.3
16	S00015	Female	17	Bachelor	Medium	Yes	Quiet	3.33	89.7	6.01	3.2		57.8
17	S00016	Male	15	High School	High	Yes	Moderate	2.47	95.3	5.67	0		83
18	S00017	Female	17	High School	Medium	Yes	Moderate	4.25	89.1	8.03	2.2		96.8
19	S00018	Female	17	High School	High	Yes	Quiet	2.26	99.5	6.3	2.3		70.3
20	S00019	Female	18	High School	Medium	Yes	Quiet	1.09	67.9	7.63	1.1		91.8
21	S00020	Male	17	High School	Low	Yes	Noisy	4.13	84.4	9.17	2.3		90.1
22	S00021	Female	16	High School	Medium	No	Quiet	4.59	84	7.79	3.1		100
23	S00022	Male	15	High School	Medium	Yes	Noisy	2.01	82.8	6.58	3.6		100
24	S00023	Female	17	Master	Medium	Yes	Moderate	5.35	90	6.26	2.7		88.7
25	S00024	Female	15	High School	Medium	Yes	Moderate	3.7	87.1	6.69	3.5		78.2
26	S00025	Female	15	Bachelor	Medium	Yes	Quiet	3.52	93.2	5.97	2.5		55.8
27	S00026	Female	15	High School	Medium	Yes	Moderate	2.38	73.2	5.86	0		94.6
28	S00027	Female	17	Master	Low	Yes	Quiet	4.22	100	6.2	2.3		97
29	S00028	Female	17	Master	Low	Yes	Quiet	1.17	81.1	7.53	0.7		76.1
30	S00029	Female	15	Bachelor	Low	Yes	Noisy	2.59	87.9	6.48	2.7		85.3
31	S00030	Female	16	Bachelor	Low	Yes	Quiet	2.12	82.9	6.77	2.2		64.6
32	S00031	Female	15	High School	Medium	Yes	Moderate	1.43	85.8	7.47	2.2		71
33	S00032	Male	16	PhD	Low	Yes	Noisy	4.04	79.7	7.73	1.2		60.2
34	S00033	Female	15	Master	Medium	No	Quiet	3.41	87.1	5.07	1.9		84

The dataset contains 10,000 rows and 23 columns. It includes both numerical variables and categorical variables. Numerical variables include study hours per day, attendance rate, sleep hours, subject scores, final exam score, and previous GPA. Categorical variables include gender, parental education, family income, internet access, tutoring, pass/fail status, and grade category.

The purpose of this analysis is to explore the factors that may affect students' final exam performance, such as study habits, attendance, internet access, and parental education.

2. Data Cleaning

2.1 Check missing values

Missing values were checked across all columns using Excel. No missing values were found in the dataset, so no replacement or removal was required.

2.2 Check duplicate rows

Duplicate rows were checked using the Remove Duplicates function in Excel. No duplicate rows were found in the dataset.

B. Descriptive Statistics

3.1 Table of descriptive statistics

G	H	I	J	K
Variable	Average	Minimum	Maximum	Standard Deviation
study_hours_per_day	3.0	0.5	7.2	1.2
attendance_rate	84.7	50.8	100.0	9.5
sleep_hours	7.0	4	10.0	1.0
social_media_hours	2.5	0	8.0	1.4
assignment_completion_rate	79.5	40	100.0	13.8
math_score	49.5	0	100.0	13.9
reading_score	49.8	0	100.0	13.4
writing_score	49.7	0	100.0	13.4
science_score	49.8	4.8	100.0	13.8
final_exam_score	49.7	4.4	97.8	12.1

Descriptive statistics were used to summarize the main numerical variables in the dataset, including study hours per day, attendance rate, sleep hours, social media hours, assignment completion rate, final exam score, and previous GPA.

The descriptive statistics show that the average final exam score was 49.68, with the lowest score of 4.40 and the highest score of 97.80. The average study hours per day was 3.02 hours, while the average attendance rate was 84.70%. This suggests that students generally had good attendance, but their final exam performance varied widely.

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C. Pivot Tables

The first pivot table compares the average final exam score between students with and without internet access.

Pivot Table 1: Average final exam score by internet access

internet_acce	Average of final_exam_score
No	48.92
Yes	49.77
Grand Total	49.68

Students with internet access achieved an average final exam score of 49.77, while students without internet access achieved an average score of 48.92. This suggests that internet access may slightly support student learning by helping students access online resources and complete academic tasks more easily. However, the difference is small, so internet access alone may not be the strongest factor affecting exam performance.

The second pivot table compares the average attendance rate between students who passed and students who failed.

Pivot Table 2: Average of Pass/fail by attendance rate

Row Labels	Average of attendance_rate
Fail	83.51
Pass	85.97
Grand Total	84.70

Students who passed had an average attendance rate of 85.97%, while students who failed had an average attendance rate of 83.51%. This shows that students who passed generally had higher attendance than students who failed. Therefore, regular class attendance may be an important factor in academic success.

The third pivot table compares the average final exam score between students with high study hours and students with low study hours.

Pivot Table 3: Average final exam score by study hours

Row Labels	Average of final_exam_score
High study hours	55.29
Low study hours	44.10
Grand Total	49.68

Students with high study hours achieved an average final exam score of 55.29, while students with low study hours achieved an average score of 44.10. This shows that students who spent more time studying each day tended to perform better in the final exam. Therefore, consistent study habits may have a positive relationship with academic performance.

The fourth pivot table compares students' final exam scores based on parental education level.

Pivot Table 4: Average final exam score by parental education

Row Labels	Average of final_exam_score
Bachelor	49.90
High School	49.42
Master	49.78
PhD	49.79
Grand Total	49.68

The average final exam scores were very similar across different parental education levels. Students whose parents had a Bachelor degree achieved an average score of 49.90, while students from High School, Master, and PhD backgrounds had average scores around 49.42 to 49.79.

D. Charts

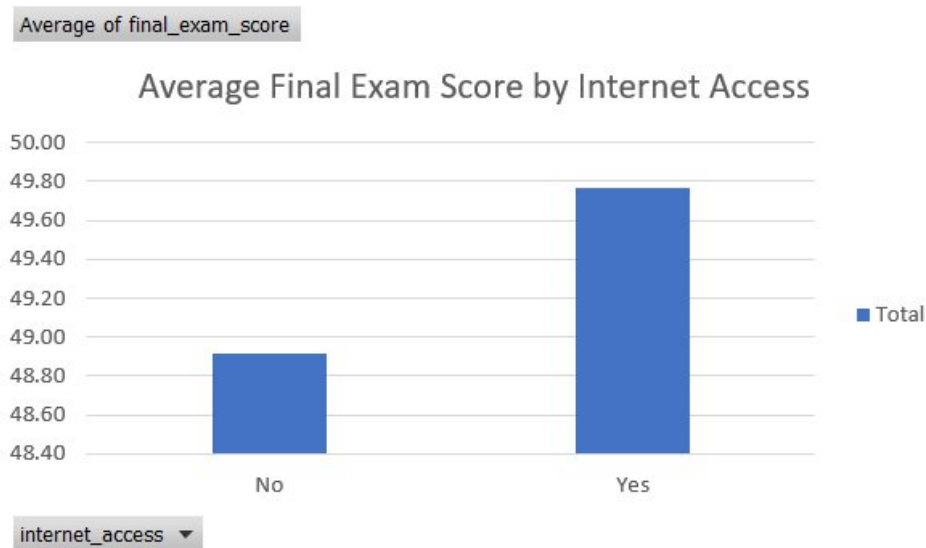


Chart 1: Average final exam score by internet access

This chart shows the difference in average final exam scores between students with and without internet access. Students with internet access achieved a slightly higher average score than students without internet access.

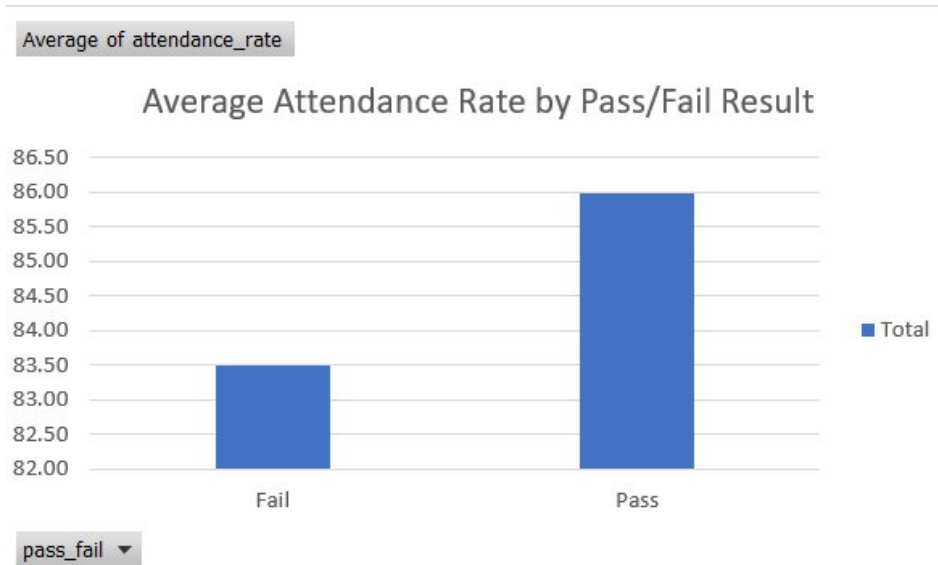


Chart 2: Average of pass/fail by attendance rate

This chart compares the average attendance rate of students who passed and students who failed. Students who passed had a higher average attendance rate, suggesting that regular attendance may support better academic performance.

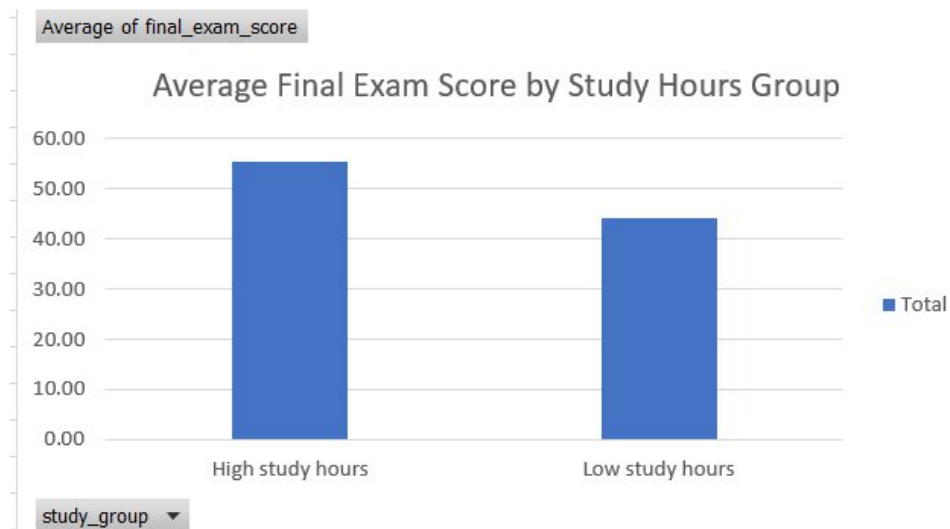


Chart 3: Study hours vs final exam score

This chart shows the relationship between study hours and final exam performance. Students with high study hours had a much higher average final exam score than students with low study hours.

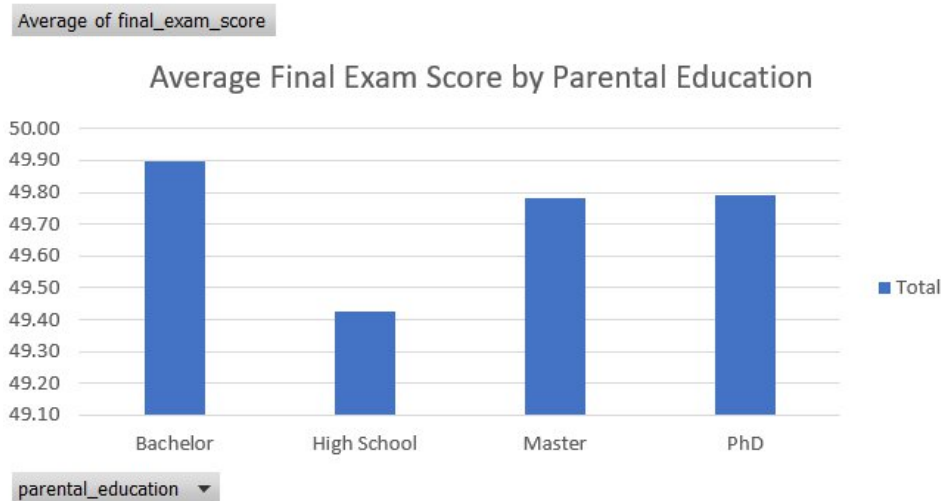


Chart 4: Average final exam score by parental education

This chart compares students' final exam scores based on parental education level. The results show that the differences between groups were small, suggesting that parental education may not be the strongest factor affecting final exam performance in this dataset. This suggests that parental education may not be a strong factor in this dataset. Study hours and attendance appear to have a clearer relationship with student performance.

E. 2 Key Insights

Insight 1: Study hours and final exam score

Students with high study hours achieved an average final exam score of 55.29, while students with low study hours achieved an average score of 44.10. This shows that students who spent more time studying each day tended to perform better in the final exam. Therefore, consistent study habits may have a positive relationship with academic success.

Insight 2: Attendance and pass/fail result

Students who passed had an average attendance rate of 85.97%, while students who failed had an average attendance rate of 83.51%. This indicates that regular class attendance may be an important factor in academic success. Students who attend classes more often are more likely to understand lessons and perform better in exams.

Insight 3: Internet access and final exam score

Students with internet access achieved an average final exam score of 49.77, while students without internet access achieved an average score of 48.92. This suggests that internet access may slightly support students' learning, but the difference is small. Therefore, internet access alone may not be the strongest factor affecting exam performance. resources.

In conclusion: Overall, the analysis shows that student exam performance is related to study habits and attendance. Students with high study hours achieved much higher final exam scores than students with low study hours. Students who passed also had a higher average attendance rate than students who failed. Internet access and parental education showed smaller differences, suggesting that personal study behavior may be a stronger factor in academic performance.

F. References

Eastern International University. (2026). *MIS 311 - Introduction to Business Analytics:*

Assignment #1 Personal Portfolio Site [Assignment brief].

Le Thi Tuyet Nhung. (2026). *Student Exam Performance dataset for MIS 311 Assignment #1*

[Unpublished Excel workbook].